

Parallel Local Alignment Algorithm for Multiple Sequences on Heterogeneous Cluster Systems

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Abstract—By taking into account communication startup overhead and the assigned processor distribution order and by applying hashing technique, a novel sequence distribution strategy is presented and the parallel local alignment algorithm for multiple sequences is designed on the heterogeneous cluster system that the computing nodes have different computing speeds and communication capabilities based on divisible load principle. The experimental results on the cluster system with heterogeneous personal computers show that, compared with the parallel algorithm with the average sequence distribution approach, the parallel local alignment algorithm for multiple sequences with the presented sequence distribution strategy can decrease the execution time of 13%~35%, and it can obtain good speedup and scalability.

Keywords—Biological information processing; Local alignment; Multiple sequences; Parallel algorithms; Heterogeneous cluster systems; Divisible loads

I. INTRODUCTION

The target of multiple sequences alignment is to find quickly the common parts among the bio-sequences in a large database. To design efficient multiple sequences alignment algorithms is an important task in Bioinformatics field such as detecting sequences in different regions for phylogenetic analysis, finding Motif and conserved regions, and structure prediction.

Generally, since there are a large number of sequences in a large genetic database, multiple sequences alignment will spend much more processing time. To design efficient and scalable parallel algorithms for sequences alignment is a very important issue. The existing parallel algorithms for multiple sequences alignment are mainly based on computing grid [1-3], shared-memory parallel random access model [4] and the homogeneous cluster systems [5,6]. For the more practical heterogeneous cluster systems that the nodes have different computing speeds, communication capabilities and sizes of main memory, this paper presents a novel sequence distribution strategy and design a parallel local alignment algorithm for multiple sequences by taking into account communication startup overhead and by applying the divisible loads principle, and evaluates the required execution time, speedup and scalability of the presented parallel local alignment

algorithm on the cluster system with heterogeneous personal computers.

II. ANALYSIS FOR LOCAL MULTI-SEQUENCE ALIGNMENT APPROACH

Basic local alignment search approach BLAST [7] provides a heuristic local alignment algorithm for multiple sequences. BLAST algorithm uses a scoring function to measure the similarity among the sequences. This score is calculated by using Karlin-Altschul algorithm. The obtained Poisson probability shows the possibility of the random similarity among the sequences.

For a given segment of the gene sequence, how to find the similar record in database and mark the position of the segment in this record is the most key part in BLAST. BLAST algorithm searches the sequence database in two-stage strategy. It first searches the important matches. Secondly, it expands the searching results on the basis of these matches. BLAST applies heuristic algorithm to search the local alignment with the highest score.

A sequence is partitioned into several short and continuous strings called words. BLAST algorithm includes the following five steps:

Step 1: length W of a string is predetermined and each word of length W in the sequence is listed.

Step 2: For every word w_i of length W in the query sequence of length n , where $1 \leq i \leq n - W + 1$, all the possible alignments are scored by a scoring matrix and the words whose score is larger than a given value of T are kept. T is selected to preserve about 50 words with high score. Therefore, there are about $50n$ words to be searched. T is a key parameter to determine the processing speed and sensitivity of multiple sequences alignment. The larger the value of T is, the faster the speed of alignment will be. The smaller the value of T is, the more remote source relations the algorithm may find.

Step 3: The algorithm searches the fields matching with some words in the word list for each sequence in the database.

Step 4: The matched fields are extended to the left and right sides, and the corresponding nucleotides are compared

until the score is lower than the given value of S . This kind of field is known as the high scoring pair. If the given value of S is smaller, BLAST algorithm will spend less time and its cost is that some correct alignments might be missed.

Step 5: A probability of random generation with high scoring pair is determined. It represents the statistical significance of the score.

Because the main strategy of BLAST algorithm is to find the majority of the matches, it can obtain the advantage of quick processing speed but it sacrifices the sensitivity. That is to say, BLAST algorithm does not guarantee that it can find a best alignment for the given query sequence and the sequence database.

III. THE PRESENTED SEQUENCE DISTRIBUTION MODEL AND PARALLEL LOCAL ALIGNMENT ALGORITHM

The divisible load principle [8] assumes that a given application can be partitioned into several subtasks with any size and these subtasks can be independently executed in parallel. To apply the divisible load principle to solve efficiently the given applications on a computation platform, the key issue is how to create the task allocation strategy to optimize the corresponding objectives. Specially, which resources should be selected to take part in computing? How to allocate the tasks to the processors according to the given order? How to determine the size of load to distribute to each processor?

To design parallel local alignment algorithm for multiple sequences on heterogeneous cluster system has the following characteristics. First, the sequence database can be partitioned into several sub-databases with different sizes, and each sequence sub-database is assigned to a computing node on the heterogeneous cluster system. Secondly, data communication only appears at the stage that the master node (master processor) distributes sequence sub-databases to the slave nodes (slave processors) and the stage that the slave nodes return the alignment results to the master node. Therefore, a sequence distribution model must be created to allocate the load with appropriate size to each slave node in order to minimize the waiting time among the slave nodes and improve the execution performance of parallel local alignment for multiple sequences.

A. Sequences Distribution Strategy

To obtain local alignment results for multiple sequences in use of the shortest time, we want to determine the size of the sequence database allocated to each slave processor, in other words, we must determine the number of the sequences which is distributed to each slave processor with different computing, communication and storage capability on the heterogeneous cluster system by computing the number of the segments in the sequences

The required processing time of parallel local alignment algorithm for multiple sequences is makes of three parts. The first one is the required time that the master processor sends the given query sequence and sequence database to each slave processor. The second one is the required time that each slave processor computes local alignment for multiple sequences. The last one is the required time is that each slave processor returns its local results to the master processor and the master processor generates the optimal local alignment results for multiple sequences.

Suppose that sequence database A has total N segments, the required time to solve optimally local alignment for multiple sequences is T_{max} , the heterogeneous cluster system is consists of $m+1$ computing nodes (processors) which they are denoted by $P_0, P_1, P_2, \dots, P_m$ respectively, where P_0 is master processor and P_i represents the i -th slave processor, $i=1\sim m$. We also assume that there is a communication link between P_0 and P_i , G_i is the start time for the communication and C_i is the required time that P_0 sends a segment to P_i in link, and E_i denotes the required time that P_i computes one segment by executing the serial local alignment algorithm, $i=1\sim m$.

We divide sequence database A into m sub-databases $A_1, A_2, A_3, \dots, A_m$. The amount of the segments in the i -th sub-database is α_i , and the i -th sub-database is allocated to slave processor P_i , $i=1\sim m$. The required time that P_0 sends the i -th sub-database to P_i is $G_i + \alpha_i C_i$, the required time that P_i compares and aligns the given query sequence with the i -th sub-database is $\alpha_i E_i$, $i=1\sim m$. To improve the utilization of the slave processors and reduce their waiting time, the problem of solving local alignment for multiple sequences on heterogeneous cluster which the nodes have different computing speed, communication ability and memory capability can be transformed into solving the following linear programming model:

$$\begin{cases} \sum_{i=1}^m \alpha_i = N \\ \sum_{j=1}^m (G_j + \alpha_j C_j) + \alpha_i E_i \leq T_{max}, 1 \leq i \leq m, \alpha_0 = 0 \\ \alpha_i > 0 \end{cases}$$

The above linear programming model can be solved by software Lingo.

B. Parallel Local Alignment Algorithm for Multiple Sequences.

The lengths of the sequences in database may be different. In order to determine the optimal sequences allocation, a continuous string with length of ST in sequences is defined as a segment, where $ST > W$ and ST is smaller than the length of the shortest sequence in database. The length of any sequence in the database is mapped into

the number of the segments. By computing the amount of the segments that should be allocated to each slave processor, the amount of the sequences and their index numbers in the sequence database can be determined and then the sequences are distributed to the corresponding slave processors.

To explain better the parallel local alignment algorithm for multiple sequences on the heterogeneous cluster system that the nodes have different computing speeds and communication capabilities and size of main memory, the procedure of distributing the segments in sequence database to each slave processor and executing parallel local alignment operation on a heterogeneous cluster with four slave processors is shown in Figure 1.

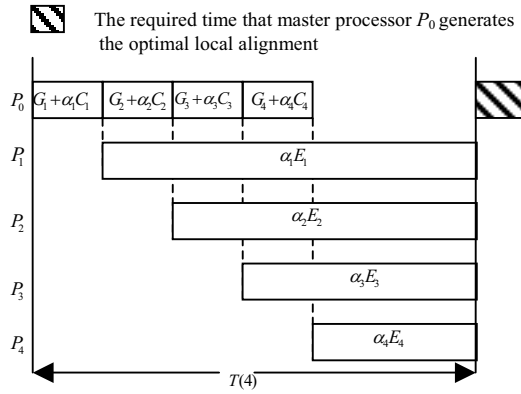


Figure 1. Procedure of segments distribution and parallel local alignment for multiple sequences on a heterogeneous cluster with 4 slave nodes

The parallel local alignment algorithm for multiple sequences with the presented sequence distribution strategy on the heterogeneous cluster system that the nodes have different computing speeds, communication capabilities and sizes of main memory includes six steps, which is described in detail as follows.

Algorithm parallel local alignment algorithm for multiple sequences on heterogeneous cluster system

Input: Query sequence Sq , Sequence database A

Output: Optimal local alignment for Sq and A .

Begin

- (1) Sequence database A is partitioned into m sub-databases $A_1, A_2, A_3, \dots$, and A_m , the amount of the sequences in sub-database A_i is denoted by sn_i , $i=1\sim m$; amount α_i of the segments in the sequences allocated to slave processor P_i is computed according to its computing speed and communication ability, $i=1\sim m$.
- (2) The amount of the sequences and their index numbers in the sequence database are determined. Master processor P_0 creates a hash table, which is

used to store the mapping relations among the segments in the sequences and sub-databases.

- (3) Query sequence Sq is first distributed to every slave processor, and then the sequences in sub-databases are distributed to the slave processors in the decreasing order of the computing speeds of the processors.
- (4) for $i=1$ to m do in parallel
- (5) for $j=1$ to sn_i do
 - (5.1) Create a pointer link list for every sequence in sub-database A_i to record the position and length of the local alignment;
 - (5.2) Slave processor P_i executes the serial local alignment algorithm to compute query sequence Sq and sub-database A_i ;
 - (5.3) Slave processor P_i returns its computed local alignment results to master processor P_0 ;
- endfor
- endfor
- (6) Master processor P_0 searches and selects the most qualified records from its received results, and re-generates the query results and reports their corresponding local alignments;

End.

IV. EXPERIMENTS

The experimental platform is a cluster system with heterogeneous personal computers connected via 100Mbps Ethernet, which each computing node runs Red Hat Linux 9.0 operating system. The heterogeneous cluster system includes four types of the computing nodes, which its detailed configuration is shown in Table I.

TABLE I. Configuration of computing nodes on the heterogeneous cluster system

Type of Nodes	CPU Frequency	Size of Memory
IBM	Pentium IV 2.0GHz	256MB
IBM	Pentium IV 2.4GHz	512MB, 1GB
HP	Pentium IV 3.0GHz	512MB

We implement the presented parallel local alignment algorithm for multiple sequences using the optimal sequence distribution strategy and the one using the average sequence distribution approach on a cluster with heterogeneous personal computers by MPI library and C programming language. By testing each communication link between the master processor and each slave processor, and each communication link between any of two slave processors several times, we obtain the required time for starting a communication and sending a character in every

link respectively. For each type of computing node, we also achieve its required unit time to compute one segment in sequences by executing repeatedly the serial local alignment algorithm for multiple sequences.

DNA sequences used in the experiments are from the United States NCBI databases [9]. For the given query sequence of 1048 characters and the sequence database with size 5.5MB, with the increase of the number of used computing nodes, we evaluate the required execution time of the presented parallel local alignment algorithm for multiple sequences using the optimal sequence distribution strategy and the required execution time of the parallel multi-sequence local alignment algorithm using the average distribution approach respectively, and their corresponding experimental result is shown in Figure 2.

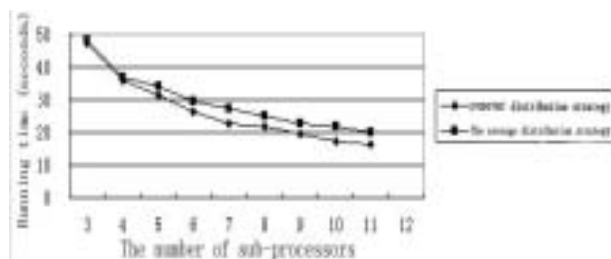


Figure 2. Execution time of the two kinds of parallel local alignment algorithms with increase of computing nodes

From Figure 2, we can see that with the increase of the number of computing nodes, both of the required execution time of the two parallel local alignment algorithms for multiple sequences using the presented sequence distribution strategy and the average sequence distribution strategy are getting shorter, but the presented parallel local alignment algorithm using the presented sequence distribution strategy is faster than the one with the average sequence distribution approach.

For the query sequence with fixed length, Figure 3 gives the required execution time of our presented parallel local alignment algorithm for multiple sequences using the presented sequence distribution strategy with the increase of computing nodes when the size of the sequence database is 1.8MB, 3.5MB and 5.5MB respectively.

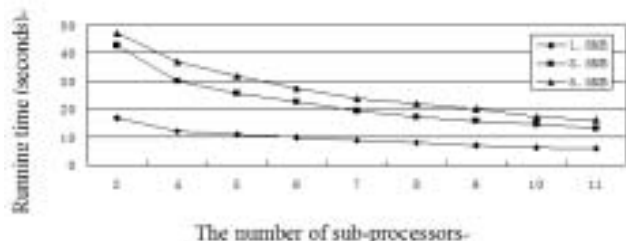


Figure 3. Execution time of parallel local alignment algorithm using the presented sequence distribution strategy with increase of computing nodes

The experimental results in Figure 3 shows that, for fixed query sequence and different size of sequence database, the

required execution time of the presented parallel local alignment algorithm for multiple sequences using the optimal sequence distribution strategy is significantly reduced when the number of used nodes is decreased.

For the fixed query sequence, Figure 4 displays the speedup of our presented parallel local alignment algorithm for multiple sequences using the optimal sequence distribution strategy with the increase of used computing nodes when the size of the sequence database is 1.8MB, 3.5MB and 5.5MB respectively.

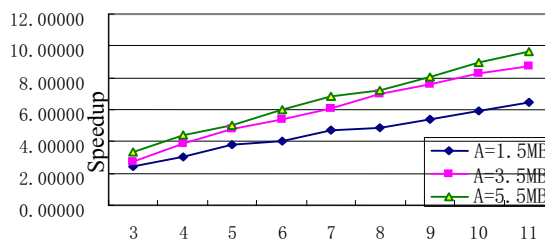


Figure 4. Speedup of parallel local alignment algorithm using presented sequences distribution strategy with increase of computing nodes

It has been seen from Figure 4 that for fixed query sequence and different size of sequence database, the speedup of the presented parallel algorithm is getting higher when the used computing nodes are increasing; furthermore, the larger the size of sequence database is, the higher speedup that the presented parallel algorithm will obtain when the amount of used computing nodes is fixed. It means that the parallel local alignment algorithm for multiple sequences using the presented sequence distribution strategy can be efficiently applied to the large-scale applications.

The scalability of parallel algorithms is one of the important issues in parallel computing discipline [10]. We evaluate the scalability of the parallel local alignment algorithm with the presented sequences distribution strategy by calculating the proportion f of the sequential execution in the parallel algorithm with the increase of the computing nodes. For the query sequence of 1048 characters and the sequence database of size 3.5MB, Table II lists the sequential ratio f in the parallel local alignment multi-sequence algorithm with the presented sequence distribution strategy when the amount of computing nodes is increasing.

TABLE II. Sequential ratio f in the presented parallel local alignment algorithm for multiple sequences

m	4	5	6	7	8
f	0.159	0.074	0.052	0.0498	0.045
m	9	10	11	12	
f	0.0362	0.035	0.033	0.032	

We can see from Table 2 that with increase of amount m of computing nodes, the value of ratio f is decreasing. Therefore, the presented parallel local alignment algorithm for multiple sequences on the heterogeneous cluster achieves good scalability.'

V. CONCLUSION

Based on divisible load principle and hashing technique, this paper presents a sequence distribution model and designs a parallel local alignment algorithm for multiple sequences on the heterogeneous cluster systems that the computing nodes have different computing speeds and communication capabilities and memory sizes by taking into account communication startup overhead and the assigned processor distribution order. The analysis and experimental results show that the presented parallel local alignment algorithm for multiple sequences obtains good speedup and scalability, and it can be efficiently applied to the large-scale applications. The next research work is to design and implement the parallel local alignment algorithm for multiple sequences on the heterogeneous cluster systems with multi-core machines.

ACKNOWLEDGMENTS

This work is supported in part by National Natural Science Foundation of China under grant NO. 60563003 and Program to Sponsor Teams for Innovation in the Construction of Talent Highlands in Guangxi Institutions of Higher Learning.

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